Some topics on derivatives and hedge accounting

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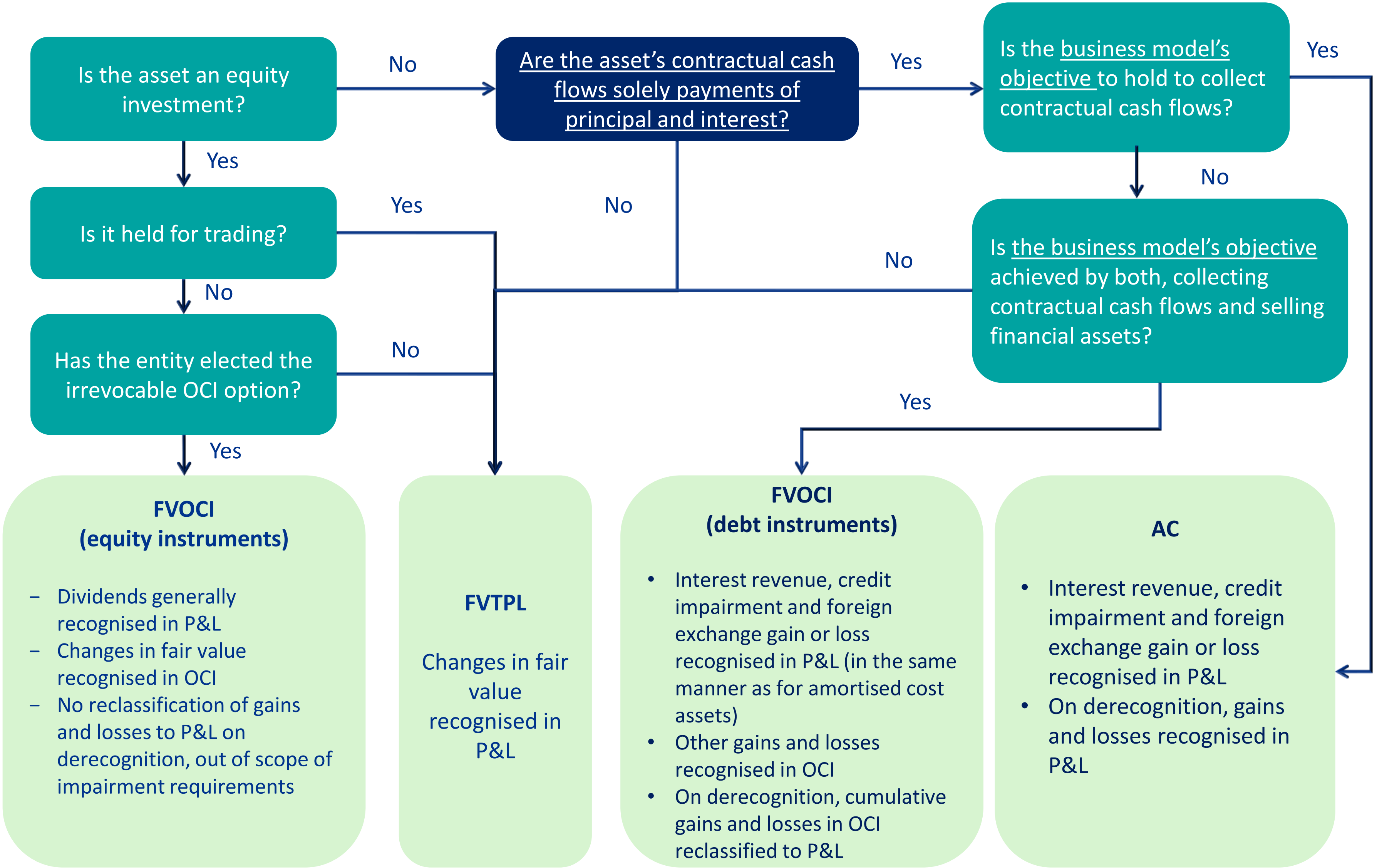
Financial assets classification under IFRS 9



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Definition of fair value (IFRS 13)

„Fair value is the price received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date.”

- Fair value is an exit price
- Fair value is a market-based measurement and focuses on market participants
- Fair value is a price in an orderly transaction
- At the measurement date

Fair value hierarchy (IFRS 13)

- IFRS 13 contains the fair value hierarchy
- The hierarchy is equal to the one introduced by IFRS 7
- IFRS 13 establishes a fair value hierarchy based on the inputs to valuation techniques used to measure fair value.

Level 1

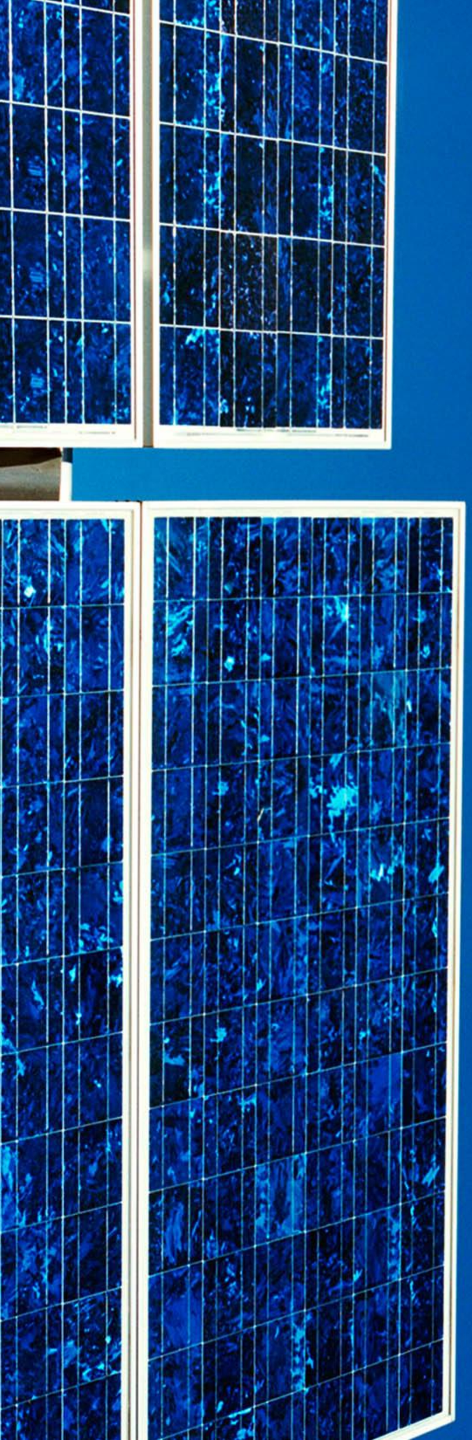
Unadjusted quoted prices in active markets for identical assets or liabilities that the entity can access at the measurment date.

Level 2

Inputs other than quoted prices included within Level 1 that are observable for the asset or liability, either directly (i.e. as prices) or indirectly (i.e. derived from prices).

Level 3

Unobservable inputs for the asset or liability.



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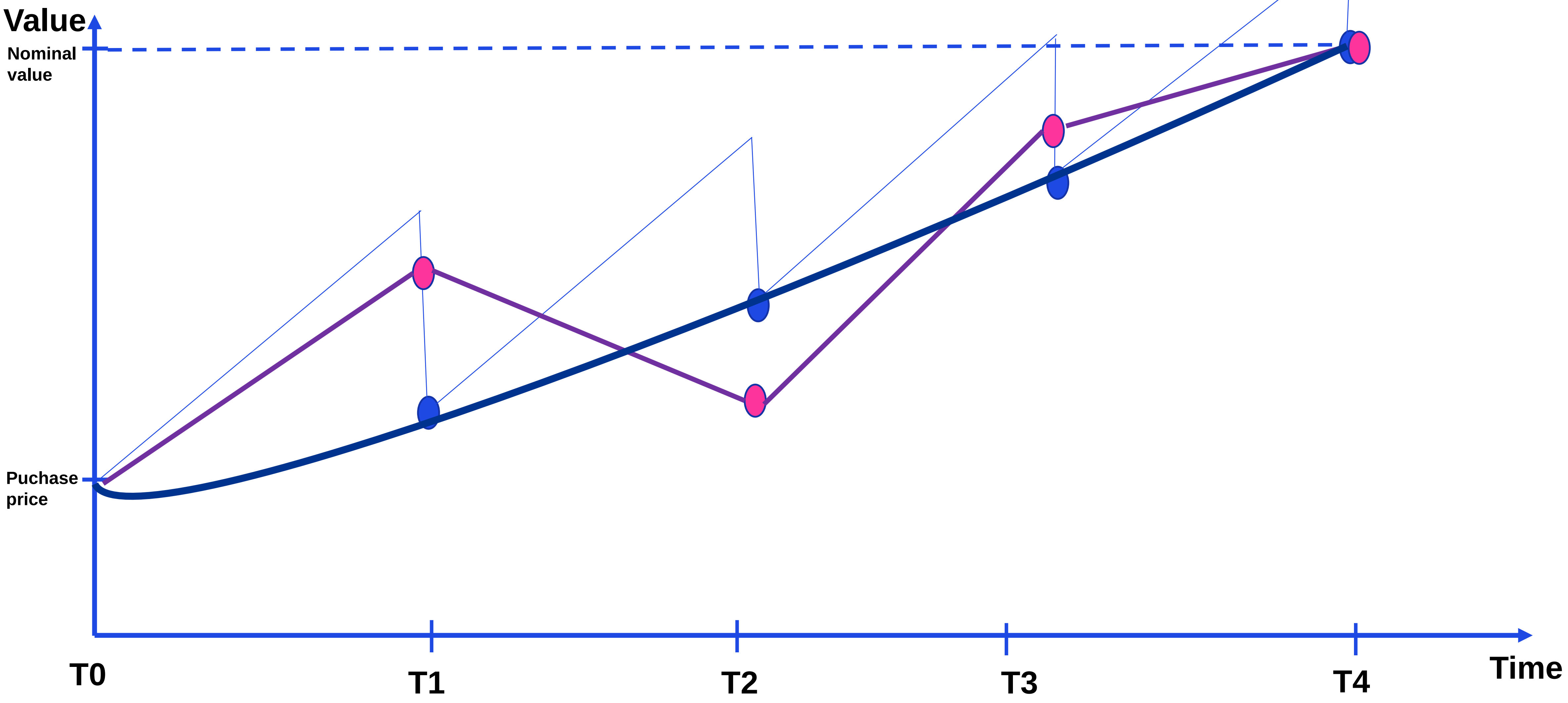
Valuation (AC, FV) examples

AC and FVOCI/FVTPL value changes - bond example (illustration)



Possible solutions to the fair value calculation

Example for loans given:

$$FV = \sum_{t=1}^{n} \frac{CF_t}{(1 + \boldsymbol{i})^t}$$

$$FV = \sum_{t=1}^{n} \frac{CF_t}{(1 + \boldsymbol{r_f} + \boldsymbol{RC} + \boldsymbol{LiqC})^t}$$

Credit risk costs

Risk free interest rate – ***from yield curve***

Liquidity costs

$$\mathrm{N} = \sum_{t=1}^{n} \frac{CF_t}{(1+\boldsymbol{r_f}+\boldsymbol{RC}+\boldsymbol{LiqC}+\lambda)^t}$$

Calibrated value

A more sofisticated method – however, it still contains a calibrated value similar to λ :

$$FV = \sum_{t=1}^{n} e^{-r(t_i)\cdot t_i} \cdot \left(N(t_i) \cdot e^{-s(.)\cdot t_i} + Int(t_i) \cdot e^{-z(.)\cdot t_i}\right)$$

The use of the formula is also possible with continuous interest

Fair value models take prepayment assumptions into account (as a part of CF_t calculation)

Derivatives

Scope exemptions for accounting purposes!

Financial instruments

„Primary” financial instruments

Trade receivables, debt securities, loans

In hedging relationships, hedging instuments are usually derivatives.

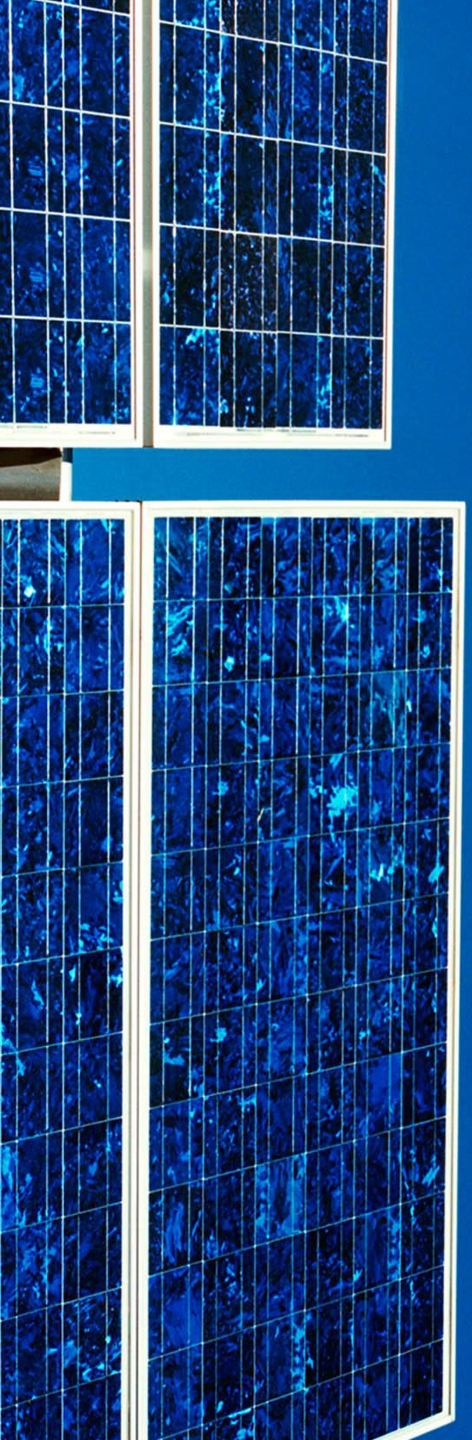
However, in certain cases hedging instruments can also be non-derivative financial instruments.

Derivatives

Forward/ futures, options, swaps, etc.

A 'derivative' is a financial instrument or other contract in the scope of IFRS 9 that has all of the following features:

- its value changes in response to one or more underlying variables - e.g. an interest rate;
- it has an initial net investment that is smaller than would be required for other instruments that have a similar response to the variable; and
- it will be settled at a future date.



Derivative valuation – real-life example

Initial recognition – IFRS 9

IFRS 9.3.1.1 An entity shall recognise a financial asset or a financial liability in its statement of financial position when, and only when, the entity becomes party to the contractual provisions of the instrument.

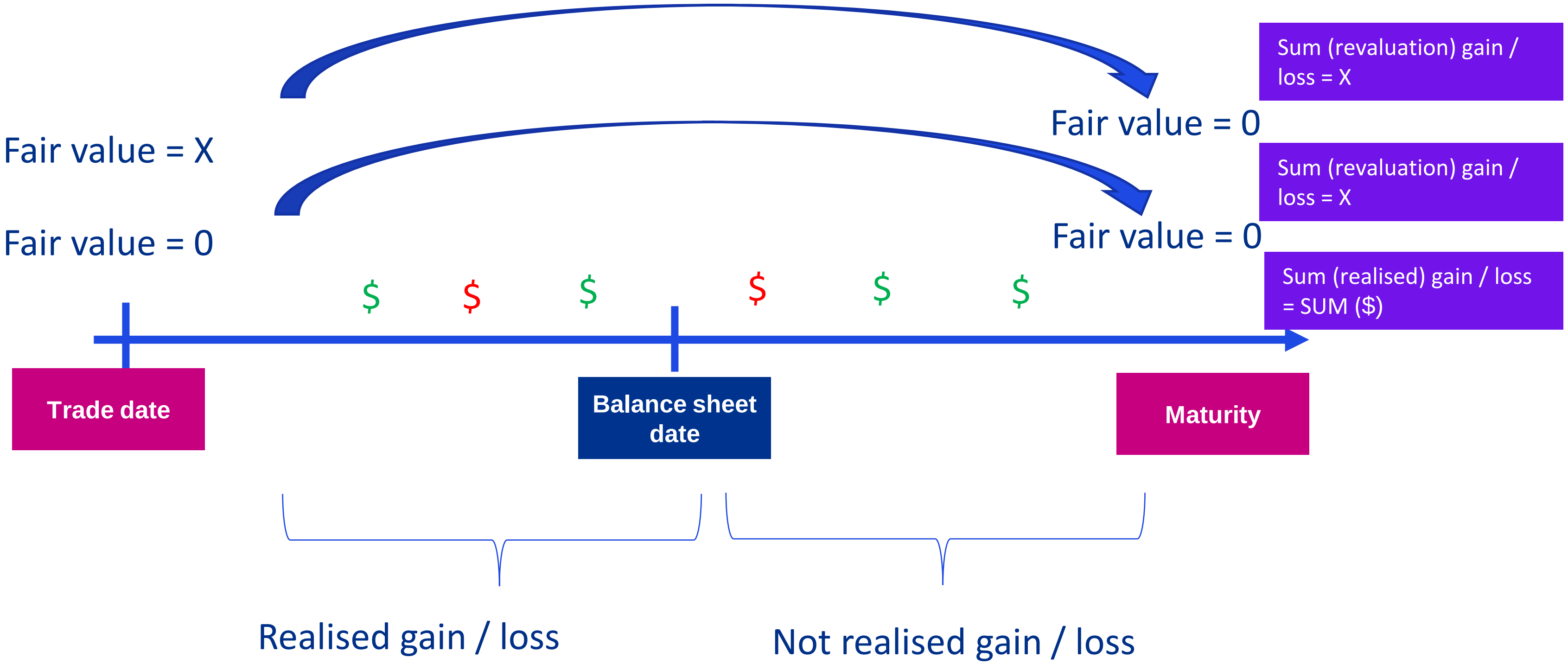
Choice between trade date accounting or settlement date accounting is not applicable.

Subsequent measurement – IFRS 9

FVTPL – Fair value through profit or loss → **Standard rule**

FVOCI – Fair value through other comprehensive income → **In case of cash flow hedge transactions or hedges of a net investment transactions (more on hedge accounting later)**

One more word on FVTPL valutaion – IFRS 9



Not realised, (fair) value of future cash-flows will affect the current fiscal year's result. It will depend on the anticipation of future cash-flows and yields (DCF).

Logic of hedge accounting

Appears in the balance sheet or NOT

Hedged risk

Value

Hedged item

Hedged risk

Value

It may

- be an asset or a liability in the balance sheet;
- transaction that are not been presented in the balance sheet

Highly probable forecast transaction

Firm commitment

Usually hedged item is not measured at FVTPL or not measured at all.

Based on the main rules

Hedging strategy (business side, not accounting!)

Hedged risk

Value

Appears in the balance sheet

Hedging instrument

Hedged risk

Value

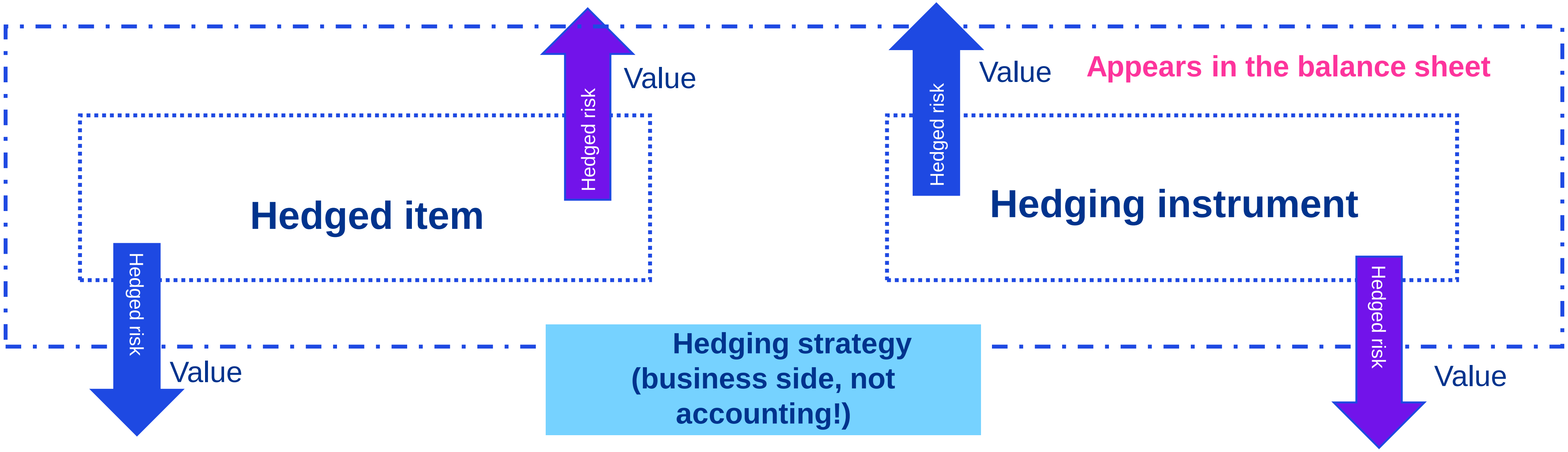
Hedging instuments **are usually derivatives**, but in some cases it van be non-derivative asset, liability (i.e. loan borrowed)

Hedge accounting = EXCEPTION RULE

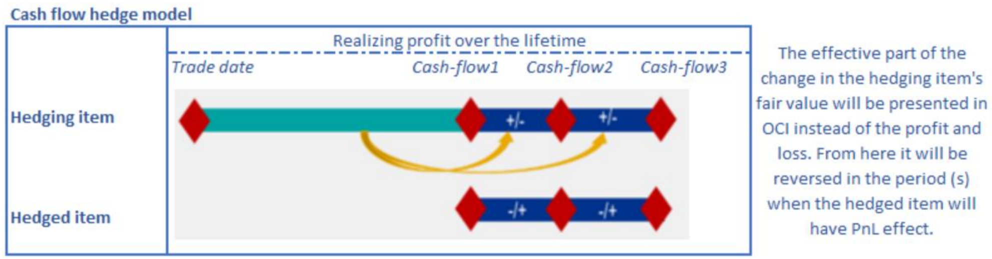
Usually hedging instrument is measured at FVTPL

Based on the main rules

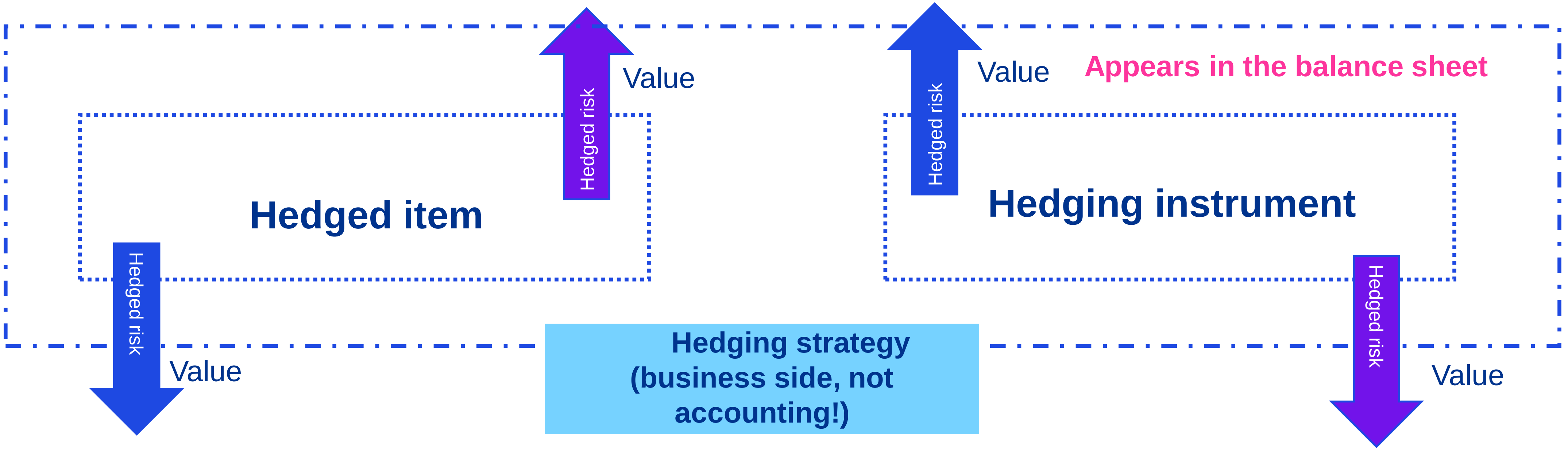
Hedging relationship – cash flow hedge solution



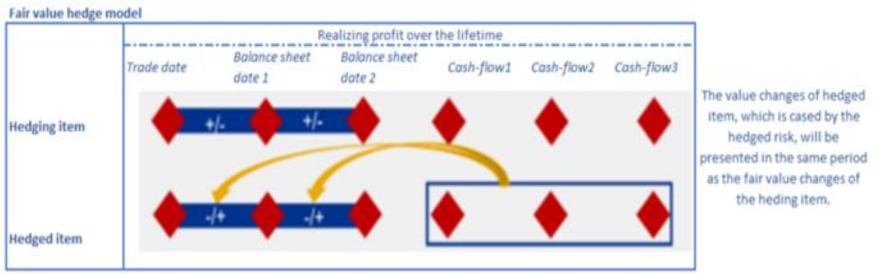
- **When the hedged risk has no balance sheet effect (i.e. interest expense, or future income effect)**
- **Hedge accounting can treat the ⚡ in a way, that it changes the main valuation rule of hedging instrument → splitting its fair value changes and „moving" a part to OCI**
- **Find the variability in the hedged item (i.e. non-fixed interest loans, variable future income)**
- **Cash-flow hedge stabilises cash-flows (arising from the hedged risk)**



Hedging relationship – fair value hedge solution



- When the hedged risk has balance sheet effect (i.e. effect on value of already existing asset, or of a firm commitment)
- Hedge accounting can treat the in a way, that it changes the main measurement rule of hedged item→ „fair valuation" to the „extent" of the hedged risk
- Find fix cash flows in the hedged item (i.e. fixed interest loan, already fixed priced contract)

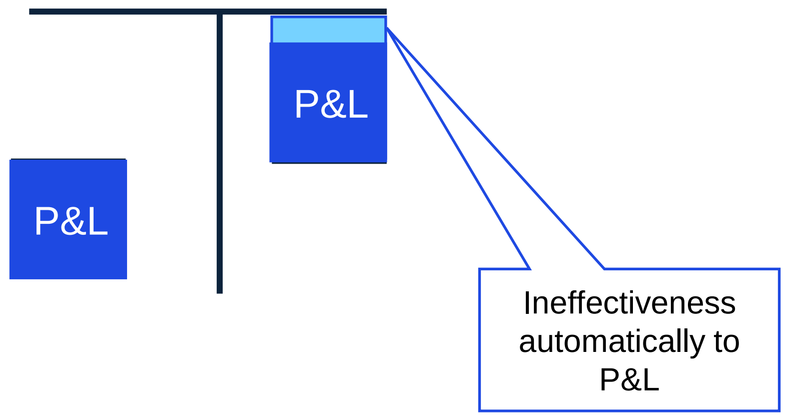


- A fair value hedging „immunize the value" of the hedged item (for the hedged risk)

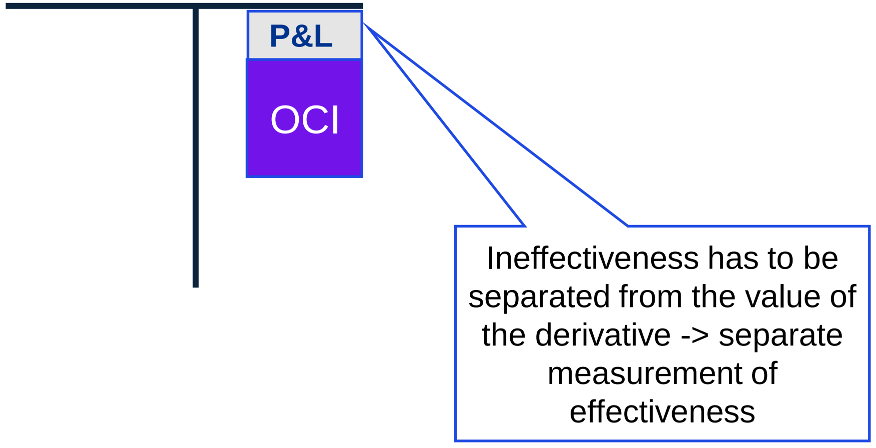
$$V = \frac{CF_1}{(1+r_1)} + \frac{CF_2}{(1+r_2)^2} + \cdots + \frac{CF_n}{(1+r_n)^n}$$

Hedging relationship – fair value vs cash-flow hedge solutions

Fair value hedge



Cash flow hedge



Hedge effectiveness

Hedge effectiveness' is the extent to which changes in the fair value or cash flows of the hedging instrument offset changes in the fair value or cash flows of the hedged item for the hedged risk(s).

As it is stated in the hedge requirements („must be effective in order to use it”)

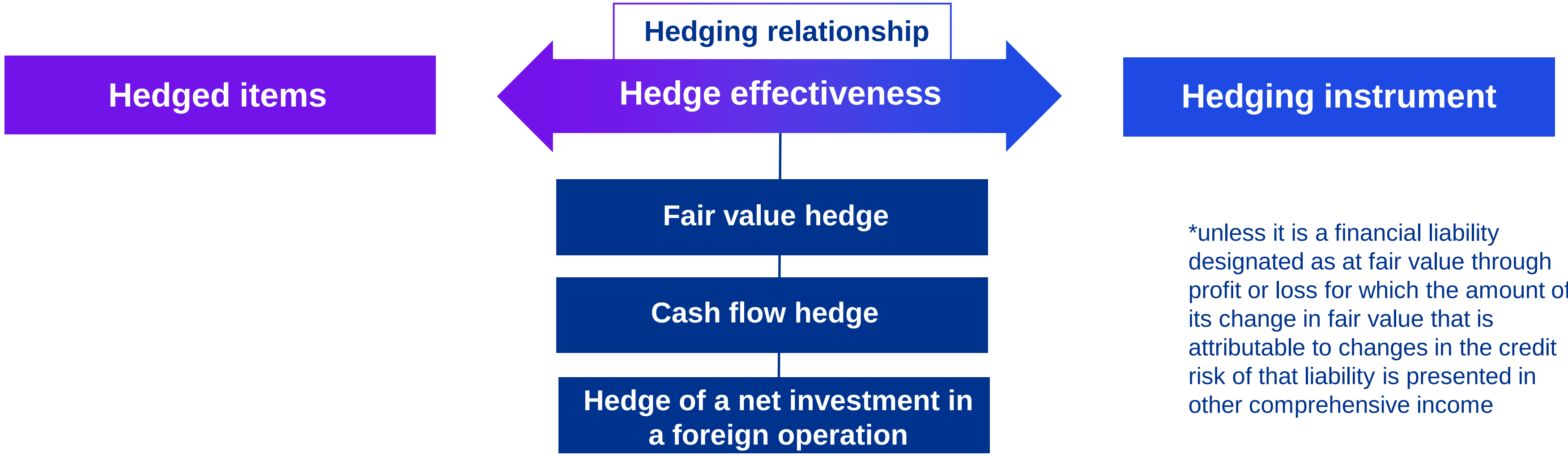
Actual accounting is also affected („account for the ineffective items”)

On hedge accounting in general

Hedged item: A hedged item can be a recognised asset or liability, an unrecognised firm commitment, a forecast transaction or a net investment in a foreign operation. The hedged item can be a single item; or a group of items. A hedged item can also be a component of such an item or group of items.

Hedging instrument: A derivative measured at fair value through profit or loss may be designated as a hedging instrument, except for some written options or a non-derivative financial asset or a non-derivative financial liability measured at fair value through profit or loss may be designated as a hedging instrument (there are exceptions*). For a hedge of foreign currency risk, the foreign currency risk component of a non-derivative financial asset or a non-derivative financial liability may be designated as a hedging instrument in some cases.

Hedge effectiveness: 'Hedge effectiveness' is the extent to which changes in the fair value or cash flows of the hedging instrument offset changes in the fair value or cash flows of the hedged item for the hedged risk(s).



*unless it is a financial liability designated as at fair value through profit or loss for which the amount of its change in fair value that is attributable to changes in the credit risk of that liability is presented in other comprehensive income

Hedge accounting - some notes

Hedge accounting is mostly about valuation. Hedged item and the hedging item is being valued. (fair valuation, present value calculation)

IFRS 13

Documentation is crucial. Always „state-the-obvious”, because it is not „obvious”. The most important is interconnection.

There are less complicated and complicated cases, a methodology must be developed, there are several decision points **(understanding of the hedging strategy is a must)**

Important to note that hedging is still possible under IAS 39 (IFRS 9.6.1.3, IFRS 9.7.2.21)

Generally, in companies’ view, applying hedge accounting results in a more realistic view, however it is worth looking how the adopter is affected. (equity/result)

There are alternatives to both fair value hedge and cash flow hedge. (fair value options)

IFRS 9.2.5, IFRS 9.4.1.5, IFRS 9.4.2.2, IFRS 9.B4.1.29-32

Introducing hedge accounting also means new applicable rules, new accounting logic and disclosures.

Useful material:

Juan Ramirez: Accounting for derivatives
Wiley 2015

Thank you for your attention!

Contact

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